Plant Biology

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Plant Biology (A08741)

Short Title:	Plant Biology
Faculty	Science and Computing
Department	Science
Cluster	Biology
Author	CDALY
Credits	5
Level:	Introductory

Description of Module / Aims

This module is designed to equip students with a basic knowledge of plant growth and physiology, including the structure and function of various parts of the plant. Students will also learn about the main living processes in the plant, and the responses of plants to environmental conditions. The module also deals with the basics of plant inheritance and biotechnology. The student will learn to operate safely in a laboratory, and perform basic laboratory experiments with plants.

Programmes

BIOL-0007	BSc (Common Entry) (WD_SSCIE_D)	1 / 2 / E
BIOL-0007	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 2 / M
BIOL-0007	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 2 / E
BIOL-0007	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 2 / E
BIOL-0007	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 2 / E
BIOL-0007	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 2 / E
BIOL-0007	BSc in Agricultural Science (WD_SAGRI_D)	1 / 2 / M
BIOL-0007	BSc in Agriculture (WD_SAGUL_D)	1 / 1 / M
BIOL-0007	BSc in Agriculture (WD_SAGUL_D)	2 / 4 / E
BIOL-0007	BSc in Food Science with Business (WD_SFDSC_D)	1 / 2 / M
BIOL-0007	BSc in Forestry (WD_SFORS_D)	1 / 1 / M
BIOL-0007	BSc in Horticulture (Botanic Gardens) (WD_SHORT_DX)	2 / 4 / E
BIOL-0007	BSc in Horticulture (Waterford-Kildalton) (WD_SHORT_D)	2 / 4 / E
BIOL-0007	BSc in Horticulture (WD_SHORT_D)	1 / 1 / M
BIOL-0007	BSc in Horticulture (WD_SHORT_DX)	1 / 1 / M
BIOL-0007	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 2 / E

Indicative Content

- Plant cell structure and function
- Plant evolution, classification and taxonomy
- Plant reproduction, genes and cell division
- Plant physiology: from embryo to establishment including anatomy of seeds, stems, roots, leaves and flowers; function of specialised stems, roots and leaves. Function of mycorrhizae on roots; secondary growth in wood; transport within plants
- Biochemical processes of plants: photosynthesis and respiration
- Plant growth and development as a function of plant growth regulators, light, water and nutrient availability

- The carbon cycle and plants' role in mitigating climate change

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Illustrate the structure and function of plant cells and plant organelles.
2. Summarise the structure, function and anatomical variability of plants.
3. Describe the major biochemical processes in plants.
4. Identify wild plants using their Latin names.
5. Operate safely in a laboratory and perform basic laboratory experiments with plants.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Online tutorials.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3
Continuous Assessment	50%	1,2,3,4,5
Lab Report	40%	1,2,3,4,5
Project	10%	4

Assessment Criteria

<40%: Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.

40%-49%: Demonstrates a limited knowledge of the subject matter. May have some problems with critical thinking and problem solving. Adequate practical skills.

50%-59%: Demonstrates satisfactory general knowledge of the main issues within the subject. Good practical skills.

60%-69%: Demonstrates skilled and sound knowledge of the subject matter. Shows ability to analyse and logically discuss in an effective way. Good critical thinking and a high level of competence and efficiency in theoretical and practical areas of this subject.

70%-100%: All of the above to an excellent level. Demonstrates authoritative handling of complex material and a highly developed and extensive understanding of the subject.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Lack, A. and D. Evans. _BIOS Instant Notes on Plant Biology_. 2nd ed. Oxford, UK. : Taylor & Francis, 2005.

Supplementary Material

- Hudson, M.J. and J.A. Bryant. _Functional Biology of Plants_. 1st ed. Oxford: Wiley-Blackwell, 2011.
- Ingram, D.S., D. Vince-Prue and P. Gregory. _Science and the Garden_. 2nd ed. Oxford: Blackwell Publishing, 2008.

Requested Resources

- Room Type: Biology Lab